

0 5 . 1

Explain what is meant by the term **phenotype**.

[2 marks]

0 5 . 2

In genetic crosses, the observed phenotypic ratios obtained in the offspring are often **not** the same as the expected ratios.

Suggest **two** reasons why.

Do **not** refer to sex-linkage or autosomal linkage in your answer.

[2 marks]

1

2

0 5 . 3

In sweet pea plants, the **F** allele for purple flowers is dominant to the **f** allele for red flowers. The **L** allele for long pollen is dominant to the **l** allele for round pollen.

A scientist carried out a genetic cross between a plant with purple flowers and long pollen (heterozygous for both genes) and a plant with red flowers and round pollen.

Table 2 shows the results of this cross.

Table 2

Phenotype of offspring	Percentage of total offspring
Purple flowers and long pollen	48.6
Red flowers and round pollen	47.5
Purple flowers and round pollen	2.1
Red flowers and long pollen	1.8

Use your knowledge of autosomal linkage to explain the results in **Table 2**.

Your answer should include:

- an explanation of what is meant by autosomal linkage
- a comment about the relative proportions of the different genotypes of gametes produced in this cross.

[3 marks]
